

BASF
We create chemistry

Care Creations™

Scalposine®

The new detox ritual to make your hair healthy

by Beauty Creations
another Care Creations™ product group Inspired by Life

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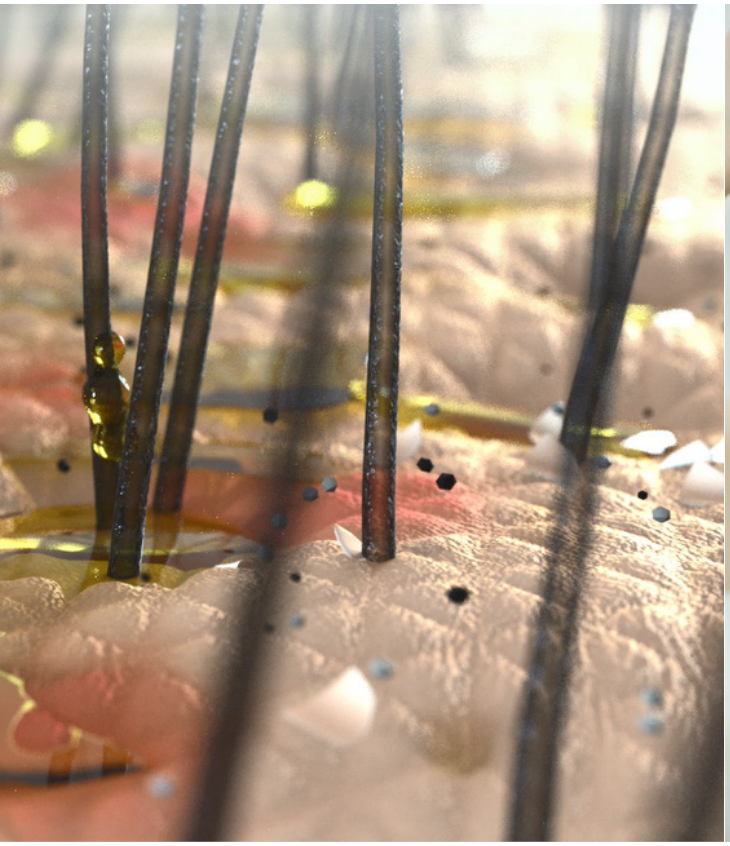
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SUMMARY FILE

Scalposine®	**BC10101**
Origin - Description	Solution of sarcosine, amino-acid
Regulatory data	
INCI	Glycerin (and) Water (and) Sarcosine
China	Each component is listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2015)
CAS#	56-81-5; 7732-18-5; 107-97-1
EINECS#	200-289-5; 231-791-2; 203-538-6
Appearance	Colorless to yellow limpid liquid
Preservative	None
Halal	Halal certified
Natural labels	None
Naturalness content (ISO 16128)	93% from natural origin
Cosmetic use	
Properties	Helps to inhibit 5α-reductase 1 Helps to reduce scalp sebum production Helps to promote oily scalp microbiota normalization Helps to reduce the scalp itchiness Helps to prevent scalp flaking Helps to improve the scalp comfort Helps to reduce the shampoo frequency
Applications	Hair and scalp care (liquid or solid shampoo, mask and conditioner) Products for oily scalp Products for sensitive scalp Products for scalp detox and purification Products for dandruff management Scalp health products
Formulation data	
Concentration of use	1%
Solubility	Soluble in water, glycerin and glycols
Incorporation method	Scalposine is incorporated during the final process at a temperature below 30°C or at room temperature for cold processing
Optimal temperature of use	15-30°C
Optimal pH	3 - 12
Patent family	Application filed





INTRODUCTION

Under normal conditions, **sebum plays a beneficial role for hair** because mixed with sweat, it forms a hydrolipidic film that protects our scalp from external aggressions and dehydration. This natural lubricant distributed along the hair shaft, protects the hair by smoothing the scales and makes the hair shiny and supple. **However, with stress and pollution, this hydrolipidic film becomes too abundant which results in a gradual asphyxiation of the scalp**. As a consequence: scalp oiliness, discomfort, itching and flakes increase...

Unhealthy scalp related to above mentioned scalp conditions are a worldwide concern that has been linked by consumer to their urban lifestyle and their environment; consumers agreed that stress and pollution play a role in the impairment of scalp health and consequently of hair health [Guinaugh O, 2019].

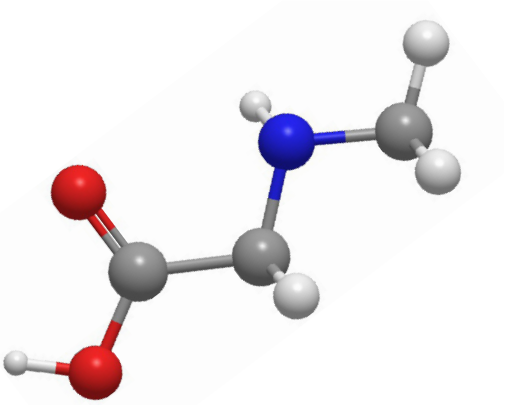
We all dream of healthy hair and a healthy scalp. Faced with daily aggression, it is tempting to bring more and more care to our hair. But this easy way of treating scalp disorders may worsen such scalp conditions, because the haircare products may create a deposit thus preventing our scalp from breathing! This is called hair build-up. Trying to correct the problem, consumers sometimes aggravate the problem.

What is less well-known is that this build-up and sebum overproduction which asphyxiate the scalp are able to modify the scalp's physiological parameters and as a result, the scalp microbiota. In fact, the scalp supports several types of microorganisms which compose a complex interdependent community living in harmony with our scalp cells. Because of the multiple environmental factors and this build-up, the fragile microbiotic balance is sometimes disturbed. Today, products for a healthy scalp mainly focus on dandruff management (using anti-fungal ingredients) and scalp microbiota is rarely considered. For BASF Beauty Care Solutions, an efficient solution to regain a healthy scalp must be holistic and should integrate microbiota scalp management because there is more and more evidence that scalp microbiota plays a large role in scalp health.

When the body is weakened by stress and pollution and clogged with impurities, detox is now a ritual chosen by many consumers to keep their body in good health. In a similar way, the scalp detox can help to get rid of impurities and rebalance the scalp. Using a natural identical amino acid Sarcosine, BASF Beauty Care Solutions has developed a new scalp detox solution which aim is:

- to **purify scalp** by decreasing oiliness and in the same time by increasing **scalp comfort** and **delaying hair regreasing time**,
- to **reset oily scalp microbiota** to recover a healthy scalp.





SARCOSINE: A NATURAL OCCURING AMINOACID WITH HEALTH AND WELLNESS BENEFITS

Description of the ingredient

Sarcosine is a natural occurring amino acid, a methylated derivative of the essential amino acid glycine found in the body. It can also be a precursor of glycine amino acid, and thus considered as taking part to the building of constitutive skin macromolecules like collagen and elastin. In the body, sarcosine comes from diet (egg yolks, legumes) or metabolization of glycine .

Health uses and biological activities

For medicinal application, sarcosine has been extensively studied for its anti-psychotic properties. It has been used to improve mental health [Strzelecki *et al,* 2016] and shows potential as antipsychotic drug as it acts as an inhibitor of glycine import into glial cells. It may reduce anxiety and increase cognitive capacities.

Sarcosine has also been proven to protect keratinocytes from osmotic stress [Graf *et al,* 2009]. Moreover, sarcosine has been reported to control skin humidity and has also been shown to inhibit ultraviolet (UV) light-induced oxidative stress in keratinocytes [Kemeny *et al,* 2007].

Interestingly, Walters and coworkers have demonstrated that sarcosine level which decreased during ageing is increased by dietary restriction [Walters *et al,* 2018]. Dietary restriction is known to attenuate decline in metabolic homeostasis during ageing. This suggests that sarcosine can fight homeostasis impairment induced by ageing; the underlying supposed mechanism is increased in autophagy which can be considered as a detoxication process [Walters *et al,* 2018]. This makes sarcosine an interesting candidate also for body detoxication.

Processing and composition

Scalposine is prepared as a sarcosine preservative-free hydroglycerine solution (Glycerin 82%; Sarcosine 6.3 - 7.7% and Water sqf. 100).

Safety / tolerability of the product

Scalposine was tested to ensure its safety under the recommended conditions of use. Scalposine does not irritate the eyes or skin and no indication of skin sensitization was observed.

Description

Scalposine is a preservative free hydroglycerine solution of Sarcosine.

INCI name: Glycerin (and) Water (and) Sarcosine

Naturalness content (according to ISO 16128): 93% from natural origin

Dose of use: 1%

China IECIC listed



Example: Illustration of a commercial sample of Scalposine, with a shampoo and a mask containing 1% of it.

** No involvement of animal genes or animal-derived substances; unlikely cross-contamination from animal substances; no animal testing has been carried out by or on behalf of BASF on the ingredients of the product after 11th March 2009 and/or after the extended deadline 11th March 2013 for the purposes of the Cosmetic Regulation (EU) 1223/2009 (statement available upon request).

SCALPOSINE - THE NEW DETOX RITUAL TO MAKE YOUR HAIR HEALTHY

Scalp, hair conditions and urban lifestyle

The scalp is a specific area of human's skin, with high follicular density and a high rate of sebum production. Often neglected because not so apparent, scalp health is getting more and more attention as it is now well admitted that hair health is also affected by common scalp disorders. The psychological impact of visible scalp problems may be very high. Unhealthy scalp related to some disorders like oily, itchy or flaking scalp is a worldwide concern that has been linked by consumer to their urban lifestyle and their environment; consumers agreed that stress, pollution and harsh cosmetics play a role in the impairment of scalp health and consequently hair health [Mintel surveys]. These scalp problems (oiliness, flaking, itchiness) affect people of different ages.

Oily scalp

Skin and scalp oiliness are linked to sebum secretion. Sebum is a lipidic secretion made of glycerides (30-50% mono, di and triglyceride), wax esters (26-30%), fatty acid (15-30%), squalene (12-20%) and cholesterol (1.5-2.5%).

Sebum production is under hormonal, insulin and insulin growth factor control. Regarding hormones, androgens stimulate the proliferation of sebocytes and the synthesis of sebum in the sebaceous follicles [Zouboulis *et al*, 1998]. Among androgens, dihydrotestosterone (DHT) plays a crucial role.

DHT level is directly determined by the specific activity of the steroid 5α-reductase 1. The enzyme steroid 5α-reductase 1 catalyzes the conversion of testosterone to the more active androgen 5α-dihydrotestosterone (DHT) which binds its receptor on sebocytes and stimulates sebum production (Figure 1). There are three isoenzymes of 5α-reductase (5α-R1, 5α-R2, and 5α-R3), which vary in different tissues and age. In adult skin, the 5α-reductase 1 (5α-R1 isoform) is expressed dominantly; sebocytes in the sebaceous follicles and epidermal keratinocytes are reported to express the 5α-R1 isoform of the enzyme [Koseki *et al*, 2015].

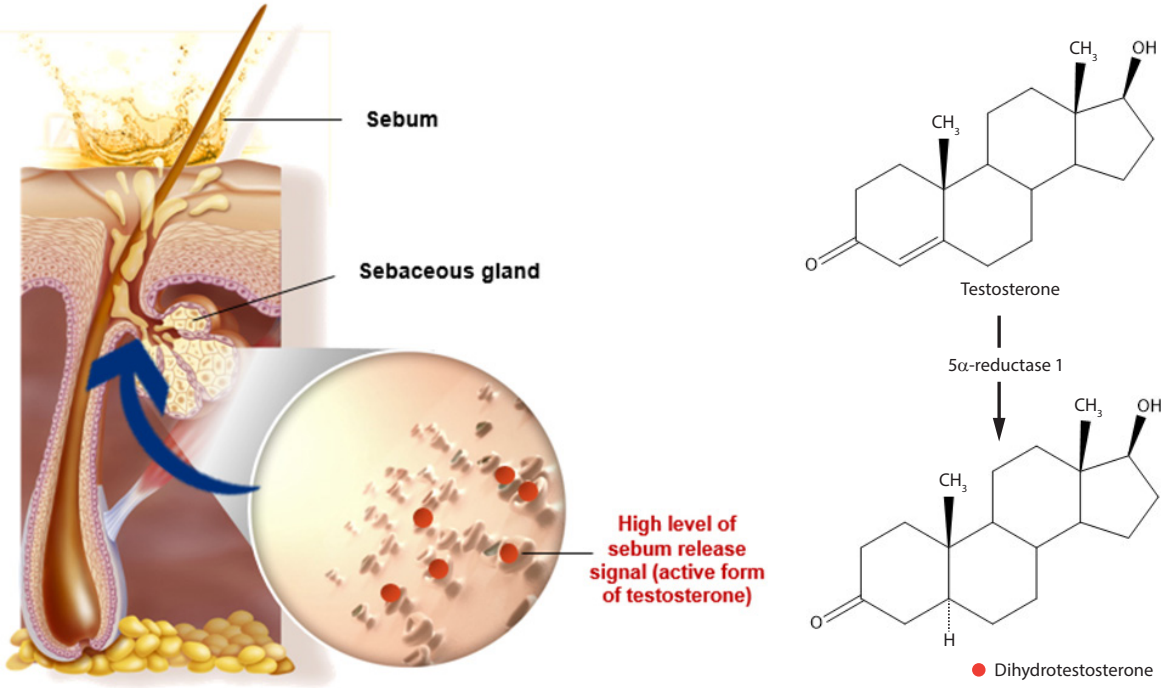


Figure 1 - Schematic representation of sebum production under DHT stimulation.

The secretion of sebum is a normal and useful phenomenon to the skin and to the hair. Sebum protects the scalp and provides shine of the hair by lubricating the cuticle. Unfortunately, hypersecretion of sebum, also called hyperseborrhoea, can become unpleasant, and causes sensations of discomfort, aesthetic disorders, or even a scalp pathology like seborrheic dermatitis. After production, sebum spreads along the hair shaft. Thus, excessive secretion of sebum can lead to greasy, flat unhealthy hair, which regreases quickly. Sebum is prone to trapping external compounds like dust or pollutants particles contributing to the feeling of a dirty scalp and hair that people tend to fix by shampooing. In four decades in Europe, the frequency of shampooing moved from once a week to 4 times a week; this frequency reach 14 times a week in some country like Korea (shampooing twice a day) [Gao *et al*, 2019]. Even showing immediate benefit on hair and scalp cleanliness, excessive shampooing can be detrimental in the long term on scalp which can become sensitive or irritated, and may create a fast regreasing.

Sebum overproduction origin may be diverse. A clinical survey conducted by Ghada linked it to psychological stress [Ghada *et al*, 2018]. Sebum is also sensitive to oxidation of its compounds like squalene and free fatty acids, which in turn creates scalp inflammation. [Gao *et al*, 2019]

Keratolytic and astringent agents are often found in care products for oily skin. Keratolytics, such as α-hydroxy acids, facilitate the elimination of impurities [Bartengev *et al*, 2011] including sebum because of their acidic nature. Astringents contract pores (pilosebaceous canal) [Son *et al*, 2013], slowing down sebum excretion. Both types of substances deal only with the consequences of hyperseborrhea and do not actually reduce sebum production. Both also have the disadvantage of having no long-lasting effect [Soirefmann *et al*, 2008].

Hyperseborrhea is a factor contributing to the development of other aesthetic scalp disorders, such as dandruff [Reygagne *et al*, 2017], and even scalp diseases. So, managing the scalp oiliness can be a way to prevent dandruff and preserve scalp health.

Scalp sensitivity

Sensitivity of the scalp may occur without necessarily being linked to a visible skin or scalp affliction; in this case an environmental stimulus causes unpleasant sensory reactions of the scalp including itching, prickling, tightness, burning and even pain. A study conducted on French population has shown that 44.2% of the population suffers from scalp sensitivity. The main triggering factors identified were pollution, heat, emotive factors, and shampoos [Godse *et al*, 2012].

Sensitive scalp showed elevated pH level, more irritated skin, and more fluorescence of porphyrins produced by propionibacteria like *P. acnes*. In a clinical study [Ma *et al*, 2018], increased sebum production was found in sensitive scalp group at occiput, among which free fatty acids, cholesteryl ester, and squalene were significantly in higher amount compared with non-sensitive scalp. Sensitive scalp also had significantly higher percentage of Propionibacterium, and lower bacterial diversity.

Scalp discomfort, like prickling and itching, have been correlated to the presence of some bacteria like Hymenobacter and Deinococcus [Park *et al*, 2017].

The choice of appropriate hair-care products represents an important aspect in the management of the sensitive scalp [Trueb *et al*, 2014].

Excessive hair loss

The scalp can also suffer from biological disturbances resulting in excessive or premature hair loss. These conditions are positively correlated with a disturbance of the scalp micro-environment and to sensitive scalp [Misery *et al*, 2008]. Excess of DHT synthesis causes miniaturization of hair by reducing anagen and increasing telogen phase leading to hair loss. Thus, some inhibitors of 5α-reductase 1 which participate in sebum production by transforming testosterone into dihydrotestosterone (DHT) can be beneficial in the prevention of hair loss [Dhariwala *et al*, 2019].

Other scalp disorders

Beside aesthetic and cosmetic concerns, scalp can also suffer from more serious afflictions linked to scalp sebum overproduction and scalp microbiota imbalance. Seborrheic dermatitis is a chronic inflammatory skin disease that affects 1-3% of the general population. This disease symptoms are close to dandruff, flaking and inflammation can be expressed. In addition, Malassezia yeasts and skin sebum levels have been described as contributing factors. In seborrheic dermatitis *S. aureus* is the most common bacteria of the skin flora [Tamer *et al*, 2018].

CHALLENGE

Urban lifestyle endangers the scalp health:
- stress leads to sebum overproduction on the scalp,
- air pollution and dust deposit on the scalp lead to build up,
- hair care products used to face these daily aggressions also deposit on the scalp and increase build up.

This contributes to an asphyxiated scalp. In addition to scalp oiliness linked to sebum overproduction, build-up can appear as well as flaking. Scalp becomes sensitive and uncomfortable. Finally, balance of scalp microbiota involved in scalp health is impaired.

Solutions on the market target generally only one of the causes of scalp asphyxia and often neglect scalp microbiota aspect. Our challenge here is to offer a holistic solution to fight scalp disorders with a detox ritual. Scalposine has been thus developed as a scalp detoxifier. Scalposine targets scalp sebum overproduction which is pivotal to numerous scalp disorders (scalp flaking, sensitivity) as well as scalp microbiome reset to a healthier normal scalp.



DEMONSTRATED EFFICACY

The ability of Scalposine, to fight scalp disorders like sebum overproduction, scalp flaking, discomfort and scalp bacteria imbalance has been evaluated *in vitro* as well as *in vivo*.
In vitro, the following properties of Scalposine have been proven:
- inhibition of 5α-reductase 1 gene expression,
- inhibition of testosterone conversion into 5α-dihydrotestosterone (DHT).
as key properties to reduce the overproduction of sebum on the scalp.

In vivo, Scalposine has proven its potential to reduce sebum production. It showed its ability to normalize scalp microbiota as well as, to increase the scalp comfort and to extend the period between two shampoos.

EFFICACY

Scalp sebum reduction - Decrease in 5α-reductase 1 gene expression

in vitro

OBJECTIVE

5α-reductase 1 catalyzes the formation of dihydrotestosterone (DHT) an androgen responsible for the stimulation of the sebocytes proliferation and the synthesis of sebum. 5α-reductase 1 is then a key target in the sebum biosynthesis [Koseki *et al*, 2015].
The aim of this study is to determine the capacity of Scalposine to reduce the expression of the 5α-reductase 1 gene in normal human keratinocytes grown in monolayers.

RESULTS & DISCUSSION

As a result of a screening of more than 100 ingredients, sarcosine was identified as the most potent ingredient to reduce the gene expression of 5α-reductase 1. Scalposine without additives (sarcosine solution) at 0.7% significantly reduced its expression by 77% ($p<0.05$) compared to the untreated control (Figure 2).

5α-reductase 1 gene expression

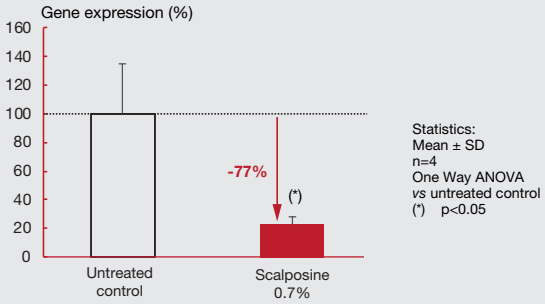


Figure 2 - Evaluation of Scalposine effect on 5α-reductase 1 gene expression in normal human keratinocytes cultured in monolayers.

CONCLUSION

Scalposine successfully inhibited 5α-reductase 1 gene expression for a decrease in transformation of testosterone in DHT and thus may help to reduce sebum overproduction in the scalp.

MATERIALS&METHODS

Cell culture

Normal human keratinocytes obtained from a facial biopsy of a 52-year-old donor were cultured in 24-well plates in defined FCS-free medium at 37°C, 5% CO_2 and 95% relative humidity until total confluence.

Treatment

Cell were incubated for 24h in culture medium FCS free and candidate to test: sarcosine at a dose equivalent to Scalposine at 0.7% and KSFM medium alone for untreated control. Each condition was tested in quadruplicate. The treatment was stopped by rinsing with PBS and discarding the medium. Keratinocytes were frozen dry before extraction.

5α-reductase 1 gene expression

Extraction of RNA

Total RNA was extracted with the «SV 96 Total RNA Isolation System» (Promega, USA). Samples were diluted in RNAse-free water to obtain a final RNA concentration of 5 ng/ml. Samples were distributed in a 96-well plate, and frozen before qPCR analysis.

Q-RT-PCR

After thawing, the reaction mixture containing SYBR Green 2x buffer, reverse transcriptase and DNA polymerase was added, as well as sense and anti-sense primers of actin genes (reporter gene) or of 5α-reductase 1 (tested gene).
50 amplification cycles in an Opticon thermocycler (MJ Research, USA) were performed.

Expression of the results

The results of test and negative control samples were first normalized to the expression of the actin gene and then expressed as percentage of the untreated control.

Statistics

The results are expressed as a mean percentage ± SD compared to the untreated control standardized to 100%. The statistical analysis was done using One-way ANOVA and Dunnett's test following Sigmaplot software recommendation (Systat Software Inc, USA). The threshold of significance was set to 5% ($p<0.05$).

EFFICACY

Scalp sebum reduction - Decrease in dihydrotestosterone (DHT) production

in vitro

OBJECTIVE

The consequence of 5α-reductase 1 gene expression reduction was tested on testosterone transformation in dihydrotestosterone (DHT) on human skin explants

Dihydrotestosterone (DHT) modulates sebaceous glands activity to enhance sebum production. The aim of this study is to determine the effect of Scalposine on the transformation of testosterone in dihydrotestosterone in explants of human skin.

RESULTS & DISCUSSION

In the conditions of this experiment, human skin explants incubated in the presence of Scalposine showed a significant reduction of transformation of testosterone into DHT. An inhibition of 36% ($p<0.05$) was obtained when Scalposine was used in culture medium at 0.14% (Figure 3). This inhibition was greater than that obtained with 1% zinc gluconate, a known inhibitor of 5α-reductase 1 [Gupta *et al*, 2014] tested in the same conditions (-19%; $p= 0.06$).

These results are consistent with those obtained in the previous study: Scalposine, inhibits expression of 5α-reductase 1 gene resulting in a decrease in synthesis of DHT, a natural stimulant of sebaceous gland activity.

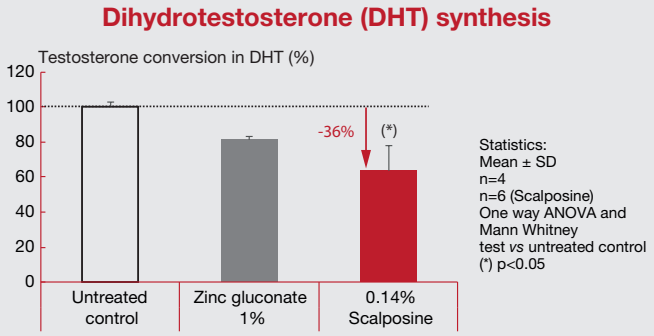


Figure 3 - Effect of Scalposine on the synthesis of dihydrotestosterone (DHT) on skin explant.

CONCLUSION

In presence of testosterone, Scalposine significantly reduced the production of DHT which plays a crucial role in the stimulation of sebaceous gland activity. This property contributes to reduce sebum overproduction in the scalp.

MATERIALS&METHODS

Cell culture

Fresh skin explants of normal human skin from a 52-year-old female patient were treated 24h, in RPMI medium (Invitrogen USA) for 24h with: Sarcosine equivalent to Scalposine at 0.14% (test in n=6), 1% of zinc gluconate (10 mg/ml, 2.2.10-2 mol/l) as benchmark (test in n=4) or culture medium alone (untreated control, n=4).

DHT Synthesis quantification

Skin explant were incubated 24h in the same previous conditions in the presence of 1, 2, 6,7-3[H] testosterone (American Radiochemical, USA). DHT contained in explants and culture media, was extracted in isopropyl ether. Skin explants were then dried and weighed. The dry residue was taken up in chloroform, and the resulting solution was then deposited on high performance thin layer chromatography plates (HPTLC, Merck, France) with reference samples of testosterone and DHT (Sigma-Aldrich, France). Elution was performed with heptane toluene/acetone-chloroform.

Visualization on chromatography plate was made possible with ammonium molybdate/sulfuric acid and heating, each spot was then scraped off the plate and assayed by liquid scintillation spectrometry (Tri-Carb 1500, Packard, USA).

Results and statistics

Results are expressed in % compared to untreated control relative to dry tissue weight.

Statistical comparison between the products was conducted with a Mann-Whitney test, and significance threshold was set at 5% ($p<0.05$).

EFFICACY

Scalp purification and reset of scalp microbiota asphyxiated by urban lifestyle

in vivo

OBJECTIVE

The aim of this study was to test Scalposine ability to reduce scalp sebum overproduction, to improve scalp comfort, to extend duration between consecutive shampoos and to shift scalp microbiota toward a healthier scalp microbiota.
Even though it is now well admitted that changes in skin and scalp environment affect microbiota composition and balance, there is not a clear description of scalp microbiota changes in case of sebum overproduction. So, we first performed at baseline a comparison analysis of the skin microbiota on oily and normal healthy scalps.
A double-blind placebo-controlled and randomized study was run on two groups of volunteers having oily scalp (lipidic index >120 using a Sebumeter). Scalposine was applied at 1% in a mask formulation three times a week for 28 days (Figure 4).

Clinical parameters, self-assessment and microbiota analysis were evaluated at the following timepoints (table 1):



Figure 4 - Clinical evaluation of Scalposine *vs* a placebo on oily scalp signs and oily scalp microbiota.

	D0	D1*	D14	D28
Sebum secretion (sebumeter)	X		X	X
Scalp purity (flakes vizualization)	X		X	X
Scalp comfort (self-assessment)		X	X	X
Regreasing time (self-assessment)	X		X	X
Microbiota analysis (bioinformatic)	X			X

Table 1 - Clinical evaluation time points.
** after first application*

RESULTS & DISCUSSION

Scalp purification

Scalp oiliness: sebum measurement

Scalposine formulated at 1% showed a fast and significant 24% reduction of scalp sebum secretion starting after 14 days of application ($p<0.001$). This effect was even improved after 28 days of application to reach a significant 31% of decrease ($p<0.001$) while the placebo showed no significant reduction of scalp oiliness at the same timepoint (Figure 5). Interestingly, Scalposine efficacy was significantly higher than those of placebo at the end of the study (-20%; $p<0.05$).

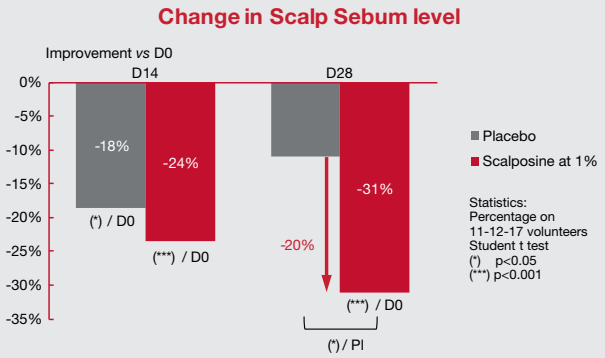


Figure 5 - Scalp sebum reduction observed after 14 and 28 days of Scalposine and placebo treatments.

Scalp purification effect: illustration

Figure 6 illustrates the scalp aspect and purity before and after Scalposine application with a visible improvement starting at 14 days and lasting until the end of the tested period.



Figure 6 - Scalp purity and flaking illustration after Scalposine application (volunteer 5NV).

In vivo

Scalp comfort: self-assessment

After the first application of Scalposine in a mask, volunteers perceived immediate benefits to their scalp. These sensorial benefits to the scalp were increasingly perceived over the study period (Table 2).

	Placebo			Scalposine		
	D1	D14	D28	D1	D14	D28
My scalp feels soothed	75%	-	-	82%*	-	-
My scalp feels more comfortable	58%	-	-	76%*	-	-
My scalp feels less itchy	-	67%	78%	-	88%*	81%*

Table 2 - Scalp soothing and comfort self-assessment. Statistics: *p-value <0.05 Two-tailed binomial test.

Scalposine provided an immediate benefit to a significant majority of volunteers, regarding scalp soothing (82%) and scalp comfort (76%) after one application (Figure 7).

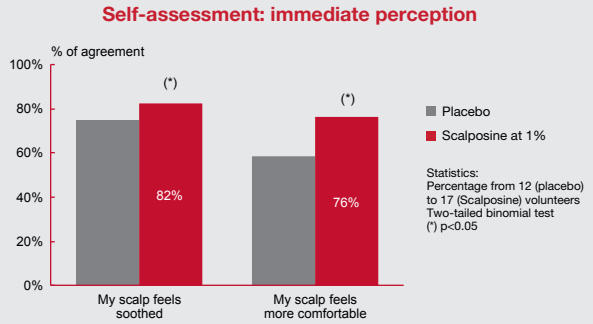


Figure 7 - Volunteers self-assessment of immediate effect of Scalposine and placebo on scalp comfort and soothing.

After 14 and 28 days of application, the improvement of scalp comfort through decrease in scalp itchiness was perceived by a significant majority of volunteers, while placebo did not provide any significant improvement according to the volunteers (Figure 8).

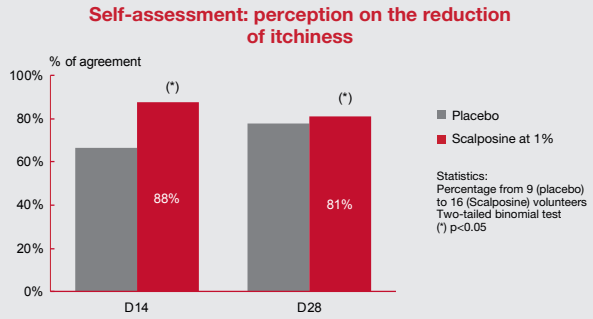


Figure 8 - Volunteers self-assessment of Scalposine and placebo effect on the reduction of scalp itchiness.

Time extension between 2 shampoos: self-assessment

People facing scalp oiliness, build-up and flaking tend to get rid of these issue by washing more frequently their hair than they should do. This may create other issues like scalp sensitiveness, irritation and fast regreasing after hair washing.

As a global solution, Scalposine showed improvement of scalp comfort and allowed to significantly extend the duration of hair regreasing and thus the time between two consecutive shampoos (Figure 9). In average, compared to the baseline after 2 weeks, Scalposine rapidly increased by 36% ($p<0.1$) the time between hair washing and noticeable regreasing as perceived by the volunteers. This effect was confirmed at the end of the study ($p<0.05$) while with the placebo mask the volunteers didn't perceive any significant effect.

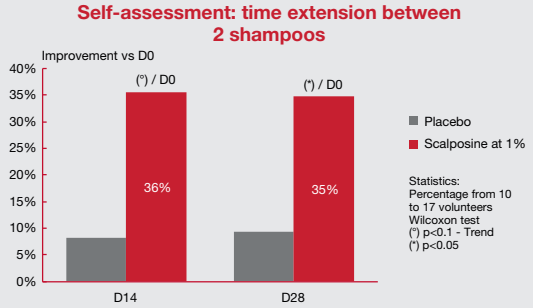


Figure 9 - Volunteers self-assessment of Scalposine and placebo effect on time between hair washing and noticeable regreasing.

In vivo

Scalp microbiome reset - Scalp microbiota analysis

Comparison of oily scalp *vs* normal scalp

After sampling at baseline, the scalp microbiota of normal healthy scalp group and the oily scalp group, shotgun sequencing and analysis revealed that there was difference in the composition of both groups (Figure 10).

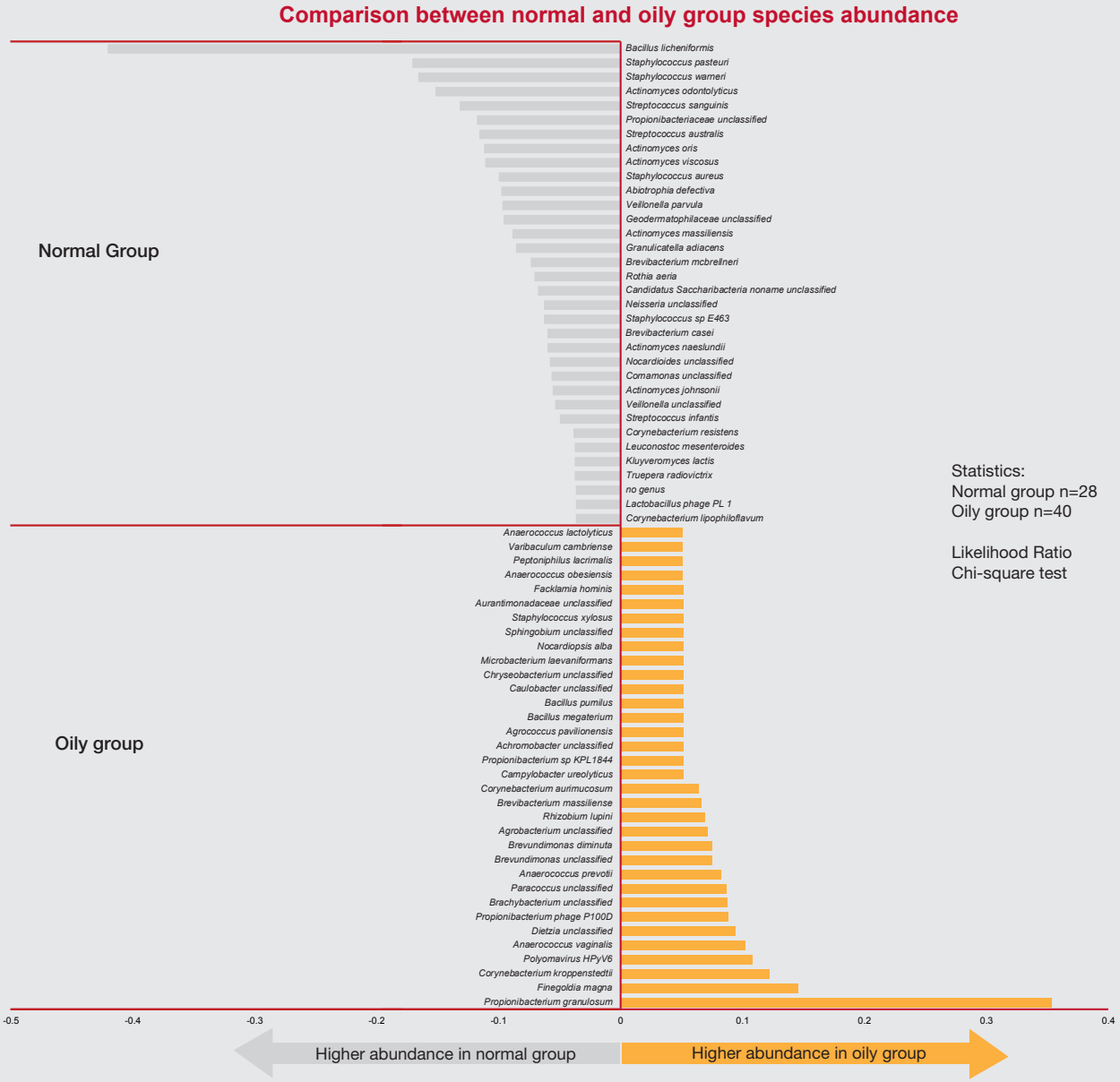


Figure 10 - Comparison of scalp microbiota composition of normal group and oily group- significant difference.

On the figure above, on x axis positive values denote significant high species abundance in the oily group while negative values denote significant high species abundance in the normal group.

Normal scalps microbiota analysis

The group with normal healthy scalp (non-oily) was characterized by a higher diversity in Staphylococcus, Streptococcus, and Actinomyces species according to species presence and absence analysis (data not shown), and by a higher abundance in Staphylococcus, Streptococcus, Actinomyces; Veillonella species and *Rothia aeria* (Figure 10). There seemed to be aerobic less lipophilic bacteria, likely due to the lower sebum levels on the normal scalps.

in vivo

Among these species of high abundance in the normal healthy scalp group, we identified 6 particular ones that should contribute to scalp health: *Staphylococcus warneri* (p< 0.01), *Staphylococcus pasteuri* (p<0.01), *Streptococcus australis* (p <0.01), *Veillonella parvula* (p<0.05), *Rothia aeria* (p<0.01) and *Actinomyces odontolyticus* (p<0.001) (Table 3).

Bacteria species	Metabolism	Potential benefits	
 *Staphylococcus pasteuri*	Gram +; aerobic	Releases AMPs against *S. aureus* (Pasteuricin)	**Inhibition of *S. aureus* pathogenicity**
 *Staphylococcus warneri*	Gram +; aerobic	Releases AMPs against *S. aureus*	
 *Streptococcus australis*	Gram +; aerobic	Antogonizes Pathogenic *Streptococcus mutans* *Lactic acid production → purification*	**Purification via short chain fatty acid**
 *Veillonella parvula*	Gram +; aerobic	Produces acetic acid from lactic acid Produces propionate → anti-dandruff, soothing	
 *Rothia aeria*	Gram +; aerobic	Uses lactic acid & produces catalase (antioxidant)	**Antioxidant**
 *Actinomyces odontolyticus*	Gram +; facultative anaerobic	Produces SOD (Super Oxide Dismutase antioxidant)	

Table 3 - Summary of potential benefits of 6 strains with significantly high abundance in normal (non-oily) group for scalp health.

Staphylococcus pasteuri and *warneri* are commensal bacteria which produce antimicrobial peptides (AMPs) to fight other pathogenic microorganisms: Pasteuricin produced by *S. pasteuri* is able to reduce *S. aureus* growth [Hong *et al*, 2018]. Warnericin another AMP is produced by *S. warneri* showed interesting properties in fighting parasites [Marchand *et al*, 2015]. *Streptococcus australis* can antagonize pathogenic *Streptococcus mutans*. [Huang *et al*, 2018].

Veillonella parvula is a commensal also found in the oral, gastrointestinal, respiratory, and female genital tract biota. It coexists with producing lactic acid bacteria like Streptococcus and Actinomyces [Long *et al*, 2018]. Veillonella uses lactic acid to produce metabolites like short chain fatty acid SCFA like propionate and acetate with interesting properties against inflammation and fungal expansion in dandruff [Kao *et al*, 2016].

Rothia aeria is able to produce catalase after lactic acid metabolization [Li *et al*, 2004].

Actinomyces odontolyticus is a facultative anaerobe known to produce a natural occurring antioxidant enzyme the superoxide dismutase (SOD) [Barkley *et al*, 1990].

In addition to that, volunteers with a normal scalp had also a high abundance of *Bacillus licheniformis* (p<0.001), other Streptococcus as well as other Actinomyces species which role on skin or scalp are not known.

Oily scalp microbiota analysis

Volunteers with an oily scalp presented a high abundance in lipophilic and anaerobic bacteria, which are hypothesized to be thriving off the sebum rich scalp (Figure 10). Most of them were Propionibacterium, Anaerococcus, Corynebacterium, Brevibacterium, Paracoccus member. We should notice in particular among the most represented species *Propionibacterium granulosum* (p<0.001), *Anaerococcus vaginalis* (p<0.01), *Finegoldia magna* (p<0.01), and *Corynebacterium kroppenstedtii* (p<0.05). These bacteria are likely to be more pathogenic than commensal:

- *P. granulosum* has been isolated from acneic skin even if its role in acne pathogenesis is not clear [Gehse *et al*, 1983; Kasprowicz *et al*, 1994],
- *Anaerococcus sp* have been reported to be involved in chronic wound [Kucisec-Tepes *et al*, 2012],
- *Finegoldia magna* has been described as an opportunistic skin pathogen [Murphy *et al*, 2014],
- *C kroppenstedtii* has been associated with extensive rosacea [Rainer *et al*, 2019].

in vivo

Effect of Scalposine on oily scalp microbiota

Species relative abundance

In the placebo group, the treatment did not induce shift in the scalp microbiota abundance. Only few significant ($p<0.05$) changes were observed at D28 *vs* D0, with a higher relative abundance in two species: *Dolosigranulum pigrum* and *Escherichia sp*, and a lower relative abundance in *Malassezia globosa* and *Dermatophilaceae* after 28 days of application (Figure 11).

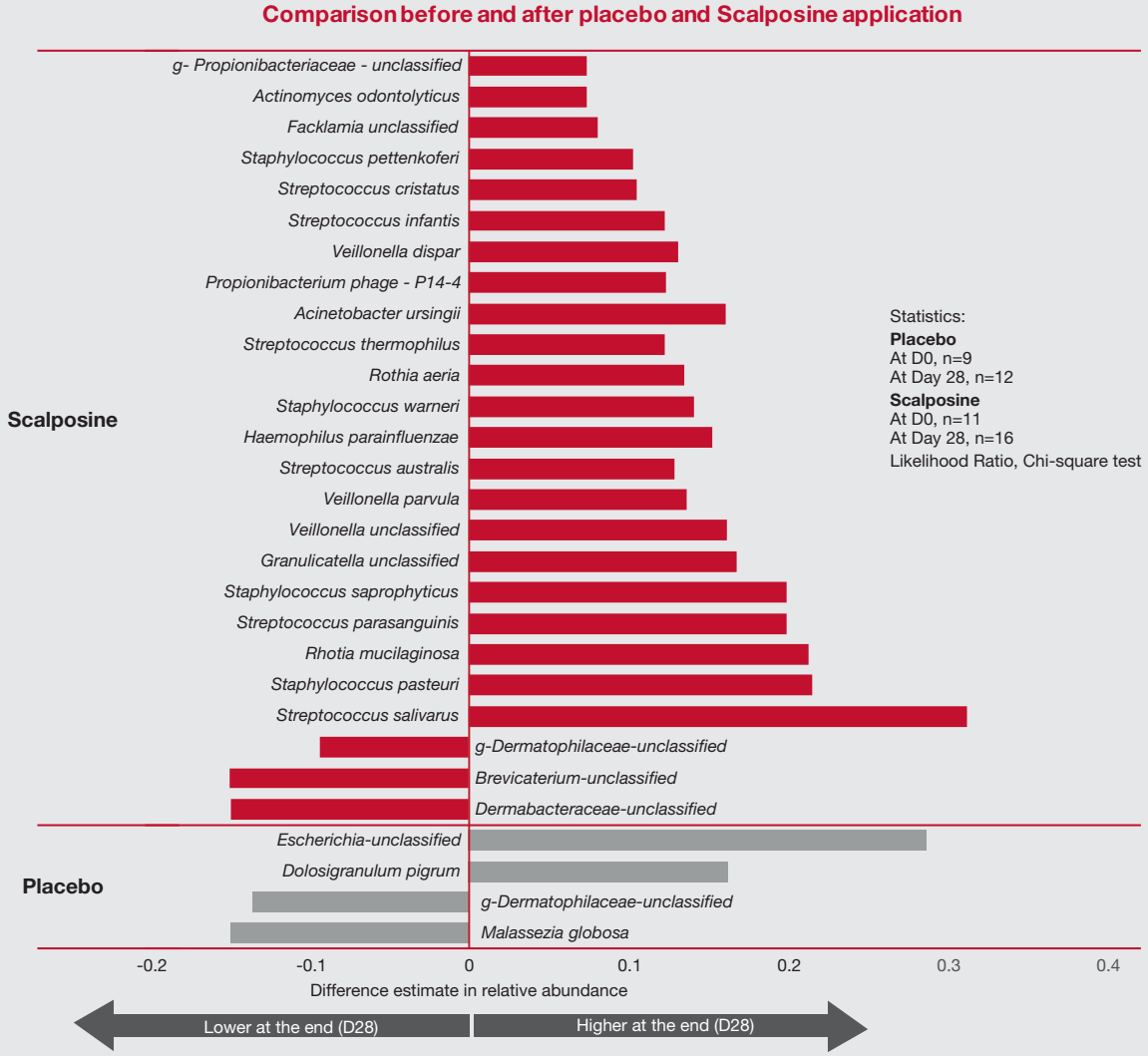


Figure 11 - Comparison of microbiome before and after placebo and Scalposine use for 28 days (significant bacteria abundance) on volunteers with oily scalp.

Negative values on the x axis denote high significant species abundance before products application compared to D28; positive values denote high significant species abundance in the groups after products application compared to baseline (D0).

On the opposite, in the Scalposine group, there was a significant shift of the scalp microbiota induced by the treatment. After 28 days of treatment, a significant increase in relative abundance of more than 20 bacteria was observed . We observed in particular the significant increase in the relative abundance of several strains, in particular those identified previously in normal healthy scalp (Figure 11, cf previous §): the 6 strains *Staphylococcus warneri* ($p< 0.05$), *Staphylococcus pasteuri* ($p<0.01$), *Streptococcus australis* ($p <0.05$), *Veillonella parvula* ($p<0.05$), *Rothia aeria* ($p<0.05$) and *Actinomyces odontolyticus* ($p<0.05$).

Furthermore, Scalposine led to a decrease in the relative abundance of three unclassified species (Dermatophylaceae, Brevibacterium and Dermabacteraceae) which were significantly dominant before product application. Overall, Scalposine treatment has shown to normalize the profile of the oily scalp microbiota into a healthier scalp microbiota.

We should highlight also that even if the relative abundance of several Staphylococcus species increased with the Scalposine treatment, this did not lead to an increase in the relative abundance of all the Staphylococcus species (overall 32% at D0 and 31% at D28, data not shown). As an increase of *Staphylococcus sp / P. acnes* ratio has been reported with dandruff occurrence [Wang *et al*, 2015], we have also checked that this ratio did not change between D28 and D0 (data not shown). Regarding fungi and particularly *Malassezia restricta* which are involved in scalp dandruff [Reygagne *et al*, 2017], their load was also not significantly modified after Scalposine treatment (data not shown).

in vivo

Scalp microbiota diversity

Unlike the placebo, Scalposine drove an increase in scalp microbiota diversity, as shown by the changes of 3 different descriptors of diversity (**A** - Taxa number. **B** -Effective specie. **C** - Shannon index) between D28 and baseline measurements.

The taxa number represents the number of organisms identified as an index for taxa richness (the representative number of species in the community). Effective species is a scaled version of Shannon index where diversity is represented using a theoretical unit of equally abundant species. Shannon Index is an alpha diversity measurement that accounts for both richness and relative abundance (evenness).

With Scalposine, the taxa number, the effective species, and Shannon index increased respectively by +53% ($p<0.05$, Figure 12A); +22% ($p<0.1$, Figure 12B), and by 36% ($p<0.1$, Figure 12C). In the placebo group, no changes were observed for these 3 diversity descriptors (+18%, -7%, -1% respectively, not significant) showing a clear trend on the microbial diversity increase for Scalposine *vs* the placebo. This increase in the microbial diversity confirms a significant shift in scalp microbiota after treatment with Scalposine.



Figure 12 - Effect of Scalposine on scalp microbiota diversity.

As sensitive scalp has been associated with excessive sebum and a lower microbial diversity [Ma *et al*, 2018], this effect of Scalposine on microbial diversity should contribute to the improvement of scalp comfort and health.

CONCLUSION

In vivo, Scalposine has demonstrated its ability to detox the scalp, i.e.:

- **to purify scalp** by decreasing oiliness (sebum overproduction) to decrease itchiness for more comfort, and to increase the hair regreasing time between two shampoos,
- **to reset oily scalp microbiota** by shifting its composition toward a normal scalp profile, with an increase in abundance of potentially beneficial species and increase in microbial diversity.

Overall Scalposine purifies the scalp and resets scalp microbiota for a healthy scalp and hair.

MATERIALS&METHODS

Study design

A double-blind randomized placebo-controlled study was conducted to evaluate the efficacy of Scalposine.

In a preliminary step, we determined the difference of the scalp microbiome profile between an oily and normal (non-oily) scalp divided in two groups.

In a second step, the effect of Scalposine applied at 1% as a mask was compared to baseline and to a placebo effect regarding the oily scalp microbiota, the sebum reduction, the scalp comfort improvement and extension of regreasing time (or time between 2 shampoos).

The clinical study was conducted in USA (Tarrytown, NY) during the month of August 2018.

Product application lasted 28 days with measurements at different time points (see Table 1, page 11).

Inclusion criteria

Volunteers from different ethnies were recruited according to their scalp sebumeter value:

Oily scalp: Sebumeter value higher than120 µg/cm^2 on their scalp.

Normal non-oily scalp: Sebumeter value between 60 and 120 µg/cm^2.

Comparison of scalp microbiota of oily *vs* normal scalps

The comparison of microbiome of oily scalp group *vs* normal (non-oily scalp group) was performed on 40 volunteers aged from 19 to 66 years old with oily scalps, and on 28 volunteers aged from 19 to 66 years old with normal (non-oily) scalps.

Comparison of Scalposine and placebo effect on oily scalps

29 volunteers applied the treatment during 28 days with either the placebo (12 volunteers) or Scalposine (17 volunteers).

Application modality

The products (a mask formulation containing Scalposine at 1% or a placebo mask, formulation in Annex 2) were applied by the volunteers themselves, at least three times a week. Products were applied for 28 days.

Approximately 20 g of Scalposine or the placebo formulation was applied on the whole scalp (dry or wet) and massaged. A cap was then put over the scalp for 15 minutes, before washing and conditioning of the hair with a provided shampoo and conditioner (Annex 2).

Evaluation method

The evaluation was conducted under temperature between 20 and 25°C and relative humidity of 35 – 45%.

Sebum secretion measurement with Sebumeter

A Sebumeter was used to measure the sebum levels of the scalps (D0, D14, D28). A special tape stretched over the measuring head was pressed to the measurement area – it was hold for 30 seconds to ensure that sebum was absorbed into the tape. Then, the measurement head was inserted into the measurement base, where the transparency of the tape is measured. The more transparent the tape is, the higher the sebum level is. A decrease in Sebumeter value reflects a decrease in scalp sebum.

Scalp visualization

The Trichoscience scope – with a 60x lens - was used to capture illustrative images of test volunteers' scalp (D0, D14, D28).

The Trichoscience scope is an image system specially designed for analysis of the scalp and hair. It consists of a camera with high levels of magnification (up to x1000) which allows it to capture detailed images of features of the hair and scalp.

Self-assessment

Self-assessment was completed under the supervision of a technician to ensure that the questionnaire is correctly filled in by the volunteer. The questionnaires were carried out on product efficacy in soothing (D1), comfort (D1) and itchiness of the scalp (D14 and D28).

For each item, the possible answers were:

- 1: agree strongly,
- 2: agree somewhat,
- 3: agree slightly,
- 4. disagree slightly,
- 5: disagree somewhat,
- 6: disagree strongly.

Time needed between 2 shampoos or regreasing time was also evaluated through questionnaire. Volunteers were asked "how long after washing your hair do you notice it becoming greasy?" (D0, D14, D28).

The possible answers were:

- 0.5: later in the day, same day as washing,
- 1: next morning, after washing,
- 1.5: later in the day, the day after washing,
- 2: two days after washing,
- 3: more than two days after washing.

Scalp microbiota analysis

Sampling procedure

A method was developed to collect microbial samples using a sterile collection swab. Scalp microbiota were sampled by creating two lines segments on the scalp of each side of the subject's head. The segments were made by separating their hair fibers with a comb to easily access the scalp for sample collection. To collect microbes, a swab head was moistened with sterile SCF-1 solution and rubbed along the scalp for 60 seconds per line segment. The swab head was then inserted into a tube containing phosphate buffer saline to release the bacteria. One swab was used to sample each scalp segment, totaling to two swabs per scalp. Both swabs were placed into the same tube containing PBS. The tubes with collected microorganisms in PBS were placed in a rack over ice until they were processed for DNA extraction.

Microbial DNA Extraction

The QIAamp DNA Microbiome Kit from Qiagen (Cat No./ID: 51704) was used to extract microbial DNA from the scalp surface and to remove residual host DNA contamination. Qiagen's supplemented protocol was followed with an exception to stopping the procedure after step 3 to freeze and store samples at -20°C until they can be later continued at step 4.

Library Preparation and Sequencing

Whole metagenome sequencing libraries were prepared from 26µl of DNA using the NEB-Next Ultra II FS DNA Library Prep kit (catalogue #E7805S). DNA was then purified, and size selected to remove excess adaptors and adaptor dimers using Ampure XP beads. Adaptor ligated DNA was then subjected to PCR amplification with universal primer and an index barcode that is unique for each sample. The PCR product was cleaned and size-selected with beads to get a 250-600 base pairs library. The library was then checked for quality and quantity on Bioanalyzer. Libraries were sent to an external service provider for paired end sequencing on the HiSeq 3000. The data was de-multiplexed, checked for quality and handed off for bioinformatics and statistical analysis.

Bioinformatics

Taxa Analysis Preparation

Contaminants, adaptors, and known artifacts were removed from the raw reads using bbduk. Quality trimming was applied to the ends of reads targeting q10 using bbduk and reads shorter than 70 base pairs after trimming were removed. Human reads were removed from the dataset by mapping reads using bowtie2 – sensitive to hg38. All other reads were processed using MetaPhlan2 to generate a taxonomy abundance profile for each sample.

Microbiota analysis- Taxonomic analysis

Presence of a species in a sample was reported when the relative abundance value of that species was above a certain threshold. This threshold value was chosen to reflect the detection limit of 10^{-6} for the distribution of species. Species prevalence was calculated based on species presence/absence of a species for compared groups. Relative abundance values were normalized prior to modelling using quantile normalization and log10 transformation. A volcano plot was used to identify differences in relative abundance between the compared subject groupings.

Statistics

Results are expressed as the mean percentage of change compared to the baseline measurement (D0).

Instrumental measurement

The statistical analysis of the evolution of the parameter as a function of time and between treatments (Scalposine or placebo) were done after analyzing of the normality distribution using Shapiro-Wilk test. The following tests were used to compare the change in the studied parameters for each product over time, as well as the differences in the studied parameters between the treatment groups:

- For those comparisons where the normality was validated, either a paired Student t-test (comparing to baseline) or a Student t-test (inter-product comparison) was used.
- For those comparisons where the normality was not validated, either a Wilcoxon Signed Rank test (comparing to baseline/D0) or a Mann-Whitney Test (inter-product comparison) was used.

Self-assessment

The efficacy evaluation analysis involves establishing frequency tables that take into account the number of responses and calculate the frequency of the different possible answers (given as percentage) to each qualitative question.

To evaluate the efficacy and the appreciation of the products for each item, two percentages Z1 and Z2 are calculated as follows:

- Z1 = favorable opinion ("Agree Strongly" + "Agree Somewhat" + "Agree Slightly"),
- Z2 = unfavorable opinion ("Disagree Strongly" + "Disagree Somewhat" + "Disagree Slightly").
- The statistical difference in frequencies (%) between favorable and unfavorable opinions is evaluated using Two-tailed binomial test at 5%.
- A Wilcoxon test was conducted to analyze the evolution of hair regreasing over time, comparing time points D14 and D28 to the baseline/D0 rating.

Scalp microbiome analysis

For diversity analysis the same rationale as instrumentation data statistical analysis were used.

For relative abundance analysis, the likelihood ratio was used to compare two competing models for species abundance: one including the term or variable we are considering (e.g. "Age", group and ethny) and one without.

The null hypothesis is that there is no improvement after adding that term to the model.

Using a chi-square test we acquire a p-value which gives the probability that this null hypothesis is true.



GENERAL CONCLUSION

Scalposine is the new detox ritual to make the hair healthy. Scalposine contributes to purify the scalp by reducing the scalp sebum, with perceptible effect on scalp comfort and extension in duration of hair regreasing. Additionally, Scalposine resets the oily scalp microbiome to a normal profile for a holistic action on oily scalp.

Healthy hair and healthy scalp are worldwide quest. But, just like our skin, our scalp is exposed to multiple factors that can impact its vitality and comfort. Thus, our scalp suffers some consequences of the surrounding pollution, stress and intensive use of hair care products that contribute to overproduction of sebum and build up deposit, making it asphyxiated and unhealthy. Finally balance of scalp microbiota involved in scalp health is impaired following scalp asphyxia.

We developed Scalposine, a nature identical amino-acid solution, as a holistic approach to treat oily scalp and discomfort and to protect it from development of unhealthy scalp conditions.

We evidenced *in vitro* that Scalposine is able to inhibit the gene expression of a key enzyme involved in sebum production (5α-reductase 1) with subsequent decrease in conversion of testosterone to dihydrotestosterone.

In a double-blind, randomized, placebo-controlled, 28-days, clinical study on volunteers with oily scalp, Scalposine showed a significant reduction of scalp sebum compared to the baseline and the placebo. It also improved scalp comfort and extend the hair regreasing time after hair washing, allowing to decrease the frequency of shampoos.

Finally, Scalposine showed a valuable potential to shift oily scalp microbiota toward a normal scalp microbiota profile by increasing the abundance of strains which are characteristics of a non-oily normal scalp, and by increasing the microbial diversity.

This double action on scalp purity and microbiota is a new detox ritual to promote a healthy scalp for healthy hair.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and Regulatory Product Information
- Composition sheet
- Specifications
- Formulation Data Sheet
- Halal statement
- Natural content (according ISO 16128)
- Vegan statement

Annex 2 - Clinical test formula

Trade name	INCI name	Placebo formulation %	Scalposine formulation %
Dehyquart F75 T	Distearoyl Hydroxy Ethylmonium Methosulfate (and) Cetearyl Alcohol	2.00	2.00
Emulgade Sucro	Sucrose Polystearate (and) Hydrogenated polyisobutene	2.00	2.00
Lanette O	Cetearyl Alcohol	4.00	4.00
Cetiol CC	Dicaprylyl Carbonate	2.00	2.00
Cetiol SB 45	Butyrospermum Parkii Butter	0.50	0.50
Euxyl PE 9010	Phenoxyethanol (and) Ethylhexylglycerin	1.00	1.00
Water	Water	85.50	84.50
Cosmedia Triple C	Polyquaternium-37 (and) Dicaprylyl Carbonate (and) Lauryl Glucoside	1.00	1.00
Dehyquart A-CA	Cetrimonium Chloride	2.00	2.00
Scalposine	Glycerin (and) Water (and) Sarcosine	-	1.00

Provided Shampoo: VO5 Normal Balancing Shampoo
Ingredients: Water, (Aqua, Eau), Sodium Lauryl Sulfate, Cocamidopropyl Betaine, Ammonium Chloride, Sodium Laureth Sulfate, Fragrance (Parfum) DMDM Hydantoin, Citric Acid, Tetrasodium EDTA, Benzyl Salicylate, Butylphenyl Methylpropional, Linalool, Yellow 6 (CI 15985), Sodium Chloride, Polysorbate 20, Tocopheryl Acetate, Panthenol, Ascorbic Acid, Niacinamide, Biotin.

Provided Conditioner: VO5 Herbal Escapes Kiwi Lime Squeeze Clarifying Conditioner
Ingredients: Water (Aqua, EAU), Cetyl Alcohol, Stearalkonium Chloride, Cetrimonium Chloride, Fragrance (Parfum), Glyceryl Stearate, DMDM Hydantoin, Hydrolyzed Collagen, Benzyl Salicylate, Disodium EDTA, Butylphenyl Methylpropional, Linalool, Hydroxyisohexyl 3-Cyclohexene Carboxaldehyde, Benzyl Benzoate, Hexyl Cinnamal, Limonene, Citronellol, Yellow 5 (CI 19140), Polysorbate 20, Blue 1 (CI 42090), Propylene Glycol, Stearyl Alcohol, Steareth-21, Tocopheryl Acetate, Rosmarinus Officinalis (Rosemary) Extract, Panthenol, Cymbopogon Schoenanthus Extract, Citrus Aurantifolia (Lime) Peel Extract, Ascorbic Acid, Actinidia Chinensis (Kiwi), Fruit Extract, Niacinamide, Biotin.

Annex 3 - Formulation examples

Advanced Scalp Detox Shampoo (HB-FR-19-BC-50777-03)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	58.30	
	Rheocare® XGN	Xanthan Gum	1.20	Rheology modifier
B	Plantacare® 2000 UP	Decyl Glucoside	14.00	Surfactant
	Plantasil® Micro	Dicaprylyl Ether, Decyl Glucoside, Glyceryl Oleate	5.00	Conditioning agent
	Plantapon® ACG HC	Sodium Cocoyl Glutamate	12.00	Surfactant
	Lamesoft® PO 65	Coco-Glucoside, Glyceryl Oleate	2.00	Conditioning agent
	Glycerin	Glycerin	3.00	Humectant
	Sodium Benzoate	Sodium Benzoate	0.50	Preservative
C	Perfume	Parfum	q.s.	Fragrance
	Scalposine™ BC10101	Glycerin, Aqua, Sarcosine	1.00	Active ingredient
D	Citric Acid (50% solution)	Citric Acid	3.00	pH Adjustment

Specifications	
pH value (23°C)	5.0
Viscosity (Brookfield; RVT; spindle 5; 50 rpm; 23°C)	2 600 mPa s

Biome Balance Mask (HB-FR-19-BC-50845-02)

Phase	Ingredients	INCI	% by weight	Function
A	Dehyquart® F 75 T	Distearoylethyl Hydroxyethylmonium Methosulfate, Cetearyl Alcohol	2.00	Emulsifier (O/W)
	Emulgade® Sucro	Sucrose Polystearate, Hydrogenated Polyisobutene	2.00	Emulsifier (O/W)
	Lanette® O	Cetearyl Alcohol	4.00	Consistency agent
	Cetiol® CC	Dicaprylyl Carbonate	2.00	Emollient
	Cetiol® SB 45	Butyrospermum Parkii Butter	0.50	Emollient
B	Water, demin.	Aqua	84.50	
	Cosmedia® Triple C	Polyquaternium-37, Dicaprylyl Carbonate, Lauryl Glucoside	1.00	Rheology modifier
C	Dehyquart® A-CA	Cetrimonium Chloride	2.00	Conditioning agent
D	Scalposine™ BC10101	Glycerin, Aqua, Sarcosine	1.00	Active ingredient
	Perfume	Parfum	q.s.	Fragrance
	Euxyl PE 9010 *(Schülke)*	Phenoxyethanol, Ethylhexylglycerin	1.00	Preservative

Specifications	
pH value (23°C)	3.9
Viscosity (Brookfield; RVT; spindle TD, Helipath; 5 rpm; 23°C)	100 000 mPa s

Disclaimer

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